the head of the staybolt and the end fullered down to the sheet, thus giving the bolt a greater holding power. I am of the opinion that this is a question that could be looked into to advantage by the Federal inspectors, and once looked into I think that there will be less steam tight bolts installed in the future.

I know of instances where several of this kind of bolt have been installed in groups. Of course the law was complied with where the broken bolts were removed, but the danger of a rupture was increased rather than diminished.

Pittsburgh, Pa. FLEX IBLE.

The Flue Question

In reading THE BOILER MAKER of late, I have noted several articles on flues—how they should be put in, what tools should be used, and how to obtain the best results after the engine has been put in service.

While talking with an Interstate Commerce man some time ago he told me of a case of how he had to start at St. Paul to eliminate the breaking in two of trains on a mountain division over 1,800 miles away. The cause of the breaking in two of the trains was that every division was allowing cars with defective draw heads and draft rigging to get by without the proper repairs. By making each division do its repair work and allowing no cars to be shipped to the next division, the breaking in two of trains on the mountain division became a rare thing.

The above will apply very well to the flue question. You may not have to go so far back to eliminate the flue trouble, but it will be some distance. For instance, a man working piece work, setting flues in an engine that is going to run on a division 700 or 1,000 miles away, cares very little if the flues last one month or one year. No boiler maker likes to put in flues steadily, especially working piece work. And, having worked piece work myself, I know that nobody suffers but the company and the flues.

I have seen the effect of piece work on flues, such defects as all flues prossered on the sheet, some only partly prossered and others not prossered at all, big beads and very small beads, coppers under the beads, and burrs left in the flue holes that should have been reamed out before the flues were put in. To improve these conditions, each division should do its own flue work when possible. Get a good, practical man on flues, give him a reasonable length of time to do a good job, and you will surely get good results.

About the filing of flues: Any boiler maker that has worked on the desert divisions knows why the scale should be removed or filed off the firebox end of the flues—if not filed they will not last long.

In putting in flues, see that all scale, dirt and oil are removed from the sheet and flue holes. Roll the coppers in good and tight. Chip the coppers off flush with the sheet, as it is very hard to keep a flue tight that has copper under the beads. All flues should be swaged and the scale filed off to insure a good, tight job. Remove the burr on the inside of the flue left by the flue cutter; if not removed it may be the cause of splitting the end of the flue when turning it over, or belling it out, and makes a poor-looking job when the flue is beaded.

The flues should be driven into the sheet good and tight to prevent stretching as much as possible. Turn the flues over enough so that the prossers will reach in far enough to set the inside bead tight up against the sheet, and at no time allow the prossers to prosser the flue on the sheet. I think the 9/16 prossers are the best for a ½-inch sheet. Drive the prosser pin in good and tight, and turn the prossers twice to insure good work. A light rolling before the flues are beaded will insure a good, tight job and make the flue perfectly round. Some men say that to roll the flue at this stage does no good, but for a test, fill the boiler with water and test before rolling the flues; then roll the flues and test again.

The distance that the flue should extend out from the sheet for a bead should be determined by the gage of the flue. Generally, 3/16 to ¼ inch will make a good average bead. When beading the flues, keep the beads rather high and do not flatten them on the sheet, as flat beads do not have the resistance to the fire and are easily collared. On marine boilers on which I have worked that had to stand a test of 325 pounds cold water they insisted on the flues being rolled after beading, to insure a tight job, and they were tight under that pressure.

In the front end, if the flues are properly shimmed and rolled tight they will never give any trouble.

THE RAMBLER.

A New Kind of Boiler Maker

Readers of THE BOILER MAKER may be interested in knowing that there are other boiler makers besides the regular kind, such as yourself.

I, too, am a boiler maker. I make them with a pointed brush, with an air brush, with india ink, and with water colors.

Many of the "photographs" of boilers which you see in THE BOILER MAKER aren't actual photographs at all, but have been made from blueprints and sometimes from rough sketches turned over to an artist like myself, and we have to "build" the boiler so well that you can "hardly tell the difference."

This may sound "fakey" to some readers, but it isn't fakey at all. Manufacturers always like to show their product as distinctly as possible with all details in place, and sometimes the only way to get around it is to have the photograph built from the ground up or to have a photograph taken and then retouched. I also do retouching, which is about as hard as making the whole boiler—so I think.

The thing that usually gives me a "pain" is the riveting job. I often wonder if real riveters feel the same way about it. I work all day painting rivets on a boiler, get an awful headache (worse even than the kind you carry around the day after), and when night comes I can't see that I have done much. It seems that there are millions of rivets in boilers. Often I get instructions like this: "The rivets in the boiler are 2 inches apart, center to center. Reduced to the size of the photograph you will therefore make them two-hundredths of an inch apart." If you don't believe it, gentle reader, just take a look at the rivets on a picture in THE BOILER MAKER, measure the distance apart, and convince yourself that I am telling the absolute truth.

Rivets, I believe, are the things that will keep me out of heaven if anything will.

Sometimes "practical men" come around, look over my shoulder, and tell me what a "cinch" I've got alongside the man who has to put up with the noise of the hammering. But I don't believe it. I believe my job of painting in rivets is the hardest in the world. I hate it. So, brother boiler makers, please pity me when you look at the pictures in your paper. They may look good, but they're mighty tedious to make.

New York. BOILER PAINTER.

It pays to file or ream the burrs in the end of a pipe, especially on small diameters. A taper reamer is better than a round file for such a purpose.

is to give such men supervision over the boiler maker and boiler work. Weighing the flogging hammer is about as fine a bonehead stunt as I have ever seen or heard of, as if a boiler maker could not strike a light or heavy blow with a six-pound hammer!

The boiler maker foreman in the roundhouse should be a good, practical boiler maker, with a number of years' experience at this kind of work. The engines should be reasonably cool, so that a man can work in them and do a good job without sweating himself nearly to death. If the engines are given the proper attention when being washed out, there will be less hot work to do.

Keep the best boiler makers in the roundhouse and eliminate the second and third class man, as a poor man in the roundhouse can do more damage in one week than several good men can make good in a month. Using the apprentice in the roundhouse is false economy, and he should only work there in the last year of his apprenticeship. The road to economy and success nowadays is in using good material, doing good work, and having reliable men to do it.

The life of the flues depends upon the material in them, how they are put in to start with, and the care that they receive from the boiler makers doing running repairs. The long stroke air hammer has ruined more flues, broken more bridges, bellied more flue sheets and cracked more flanges than any other tool, and should be abolished on hot work. The standard prosser should be used at all times, and should be long enough to set the inside bead up tight against the inside of the flue sheet. This is very important, as the inside bead acts as a brace to hold the flue tight in the sheet when the flue is expanding from the heat. Prosser the flue tight in the hole, and that is all that can be done with the prossers.

Some men claim that the smooth prosser, with the inside bead ground off, is the best. For a test take two engines with flues about the same age, running under the same conditions; use the smooth prosser on one engine and the standard prosser on the other. If the flues are giving any trouble, it will not take long to find out what prosser to use.

The flue rollers should never be used to work flues within the firebox end. They only tighten the flues on the outside edge, bell out the flues and roll them out thin, ruin the coppers, bell out and ruin the holes in the flue sheet, and if used continually will soon shear the heads off. The mandrel pin acts about the same as the rolls, and should never be used to work flues in the roundhouse. When beading the flues, use a tool that fits the bead. I prefer to use the square heel beading tool, as it is almost impossible to cut or scar the flue sheet or to work a burr into the flue hole. Do not calk the life out of the beads. A set of flues with the beads flattened down, collars or rags on the beads, the flue sheet scarred and cut, is plain evidence that some one is not attending to business.

When the fire has burned the life out of the flues and the beads become soft and mushy and will not stay tight when prossered, it is time to take the flues out and avoid engine failures. A few engine failures add considerably to the cost of a set of flues.

Some time ago I made a trip on an oil-burning locomotive over a mountain grade. When we started the flues were perfectly dry. In going 35 miles, up hill and down hill, the fireman lost his fire four times; when we arrived at the terminal, about half of the flues were leaking, and it is a wonder that all of them were not leaking. If this engine had failed on the road, the engineer would have blamed the boiler maker, as usual.

Not long ago we received an engine just out of the back shop, with new flues. All of the flues were prossered on the flue sheet. This engine, after being in service a short time, came in one morning perfectly dry, remained in the house all day, was fired up at night and moved to another track outside the house. The next morning the engine was dead and all but a few flues were leaking. This was caused by abuse from the roundhouse force. But if the flues had been properly prossered and the inside bead set tight up against the inside of flue sheet, this never would have happened, especially with a new set of flues.

If this engine had died at the other terminal, where there was nobody but an engine watchman, it would have meant the calling of a new crew and another engine, the dead-heading of the other crew back, and hauling the other engine back or sending the boiler maker over to calk the flues, costing the company in all about $25, and the boiler maker would have had a hard time explaining to the satisfaction of the master mechanic why this engine failure occurred. THE RAMBLER.

"What Constitutes Proof" and "What Kind of Boilers are the Best?"

The above excerpts are taken from the February issue of THE BOILER MAKER.

Now one might make the abrupt statement and be correct that all types of boilers are good, but some are better than others. Our friend, Mr. N. G. Near, I see from a previous article, is a college graduate—this may account for some of the statements he makes.

Getting back to the above subject, the first thing to consider is what service is the boiler expected to perform—conditions, location and the water to be used are all very essential. The opinions of mechanical men are at great variance when it comes to advising one in the purchase or choice of boiler units. Some put up the argument of having the only safety boiler, others that their boiler is the ideal of efficiency. Some advocate the horizontal baffle and some the vertical baffle. Designers choosing the horizontal baffle claim that at no time they force the gases to descend, while on the other hand we have the vertical baffle inducing the gases to ascend and descend, which has been proven to give higher efficiency and more uniform temperature.

The argument of keeping the gases longer in contact with the tube is a poor one and a detriment to the tube, and would inevitably lead to trouble. A liberal flow of gases is to be advocated at all times.

Sometimes the argument is put up by some builders that their boiler is so arranged that they mechanically assist themselves in keeping clean—provision being made for separating the solids when precipitation takes place and the solids taken care of by frequent blowing down. Whether this is so or not has not been determined.

You will find, Mr. Near, that in various parts of the country various types of boilers are considered the best in that particular locality. The best advice that I know of for the prospective buyer is to buy the best unit in the boiler market to meet the requirements. I will not discuss Mr. Near's paragraph about welding, but I will say that his seamless, rivetless boiler in the near future sounds very much like, "Did you ever see a cowless milk wagon?"

Boilers of all types have all been tested out and all their good points catalogued for commercial purposes, and these data are very elaborate in detail, as outside of patent units we know very nearly what we can expect of any type of boiler on the market at the present time. But as long as we build boilers either by the welding or riveting process, there will always be a scamp job crop up once in a while. For my part I'll take the riveted job.